③ Universe :

$U = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23\}$

Himpunan A :

$A = \{3, 5, 7, 10, 13, 16, 18\}$

Himpunan B : $B = \{6, 8, 10, 13, 15, 21\}$

Himpunan C : $C = \{4, 10, 13, 14, 20, 23\}$

a). Hit. manual

Gabungkan semua : $\{3, 4, 5, 6, 7, 8, 10, 13, 14, 15, 16, 18, 20, 21, 23\}$

Jumlah : $|A \cup B \cup C| = 15$

2. $A \cap B \cap C$

yg sama disemua : $\{10, 13\}$

$|A \cap B \cap C| = 2$

3. $(A \cup B) \cap C$

C' = selain C

$C' = U - C = \{3, 5, 6, 7, 8, 9, 11, 12, 15, 16, 17, 18, 19, 21, 22\}$

$A \cup B = \{3, 5, 6, 7, 8, 10, 13, 15, 16, 18, 21\}$

Iris = $\{3, 5, 6, 7, 8, 15, 16, 18, 21\}$

Jumlah = 9

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4. $A \ominus B$ (simetris difference)

$(A \cup B) - (A \cap B)$

$A \cap B$: $\{10, 13\}$

Hasil : $\{3, 5, 6, 7, 8, 15, 16, 18, 21\}$

Jumlah : 9

b) <u>Wetip mackab</u>

c) komposisi fungsi

Fungsi : $f(x) = 4x + 3$

$g(x) = \frac{x^2}{4} + 6$

$(f \cdot g)(x)$

$f(g(x)) = 4\left(\frac{x^2}{4}\right) + 3$

$= x^2 + 3$

$(g \cdot f)(x)$

$g(f(x)) = \frac{(4x+3)^2}{21}$

$= \frac{16x^2 + 24x + 9}{4}$

$= 4x^2 + 6x + 2.25$

d) Inverse f

$y = 4x + 3$

$x = \frac{y-3}{4}$

$f^{-1}(x) = \frac{x-3}{4}$

Pendapatan : $4 \cdot 100 + 3 = 403$

$x = \frac{403 - 3}{4} = 100$

e) klasifikasi

- linear
- gradien $\neq 0$

$f(x) = 4x + 3$

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